DARTMOUTH

THOM CHAFFEE, PHD

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EDUCATION

Stanford - PhD Geophysics, 2020-2025

Leland Stanford Junior University (Stanford), Stanford, CA

- Research on lunar paleomagnetism with Prof. Sonia Tikoo. 1 publication, 3 in prep, 9 conference talks.
- Coursework focused on planetary science, geophysical modeling, and applied mathematics.
- National Science Foundation Graduate Research Fellow, 2020-2025.
- Extracurricular focus on outreach and education including high school and undergraduate mentoring.

Caltech - B.S. Geology, Minor: Environmental Science and Engineering, 2014-2018

California Institute of Technology (Caltech), Pasadena, CA

- 3 years of geophysical lab research with Prof. Joe Kirschvink. 1 publication and 2 conference talks.

TEACHING

Lecturer, Dartmouth Earth and Planetary Sciences, 2025-Ongoing, 20 hrs/week during Winter quarters

Instructor, *Evolution of Earth and Life,* Winter 2027 (upcoming). Modified previous instructors' curricula to integrate active learning in flagship introductory lecture course. Developing original lectures, activities, and coursework. Worked with Dartmouth Center for Advancement of Learning to integrate undergraduate Learning Fellows. Will manage >10 member teaching team. 75+ students.

Instructor, *Mineralogy*, Winter 2026. Developed original curricula & delivered instruction including laboratory and field. Trained upper-level majors on petrographic microscope, hand sample analysis, scanning electron microscopy, and X-ray diffraction. Supervised 1 graduate Teaching Assistant. 4 students.

Guest lecturer, various institutions:

Paleomagnetism and rock magnetism course, Kent State University, 16 April 2026

Paleomagnetism course, Stanford Geophysics, 18 Nov 2025

60 minute invited guest lectures on micromagnetic modeling, including 20-minute demonstration and 5-10 minutes of student Q&A.

Formation and Dynamics of Planets course, Stanford Earth and Planetary Sciences, Nov 2024

90 minute invited guest lecture on planetary formation and core differentiation for a graduate course with a focus on active learning techniques.

Associate, Center for the Integration of Research, Teaching, and Learning, 2025

Certification in undergraduate education through the multi-institutional CIRTL Network, including coursework on STEM pedagogy at Stanford Center for Teaching and Learning.

Participant, Stanford Preparing Future Teaching Professors, 2025

Shadowed faculty at California State University, East Bay, to prepare for a teaching-track academic career.

Teaching Assistant, Stanford Earth & Planetary Sciences, "Introduction to Geology", Spring 2024. 15 students

Ran and graded weekly lab sections and assisted with field exercises and exam grading. 20 hrs/wk.

Teaching Assistant, Stanford Geophysics, "Introduction to Planetary Science", Spring 2023. 10 students

Gave lectures, graded problem sets, ran weekly office hours. 20 hrs/wk.

Volunteer Teacher, Stanford GeoKids, 2022-2025. 40-80 students

Run weekly Earth science educational activities for 3 classrooms of 2nd-4th graders. 5 hrs/wk.

Graduate student peer mentor, Stanford Geophysics, 2021-2025. 1 student / year

Provide mentorship and assistance to new graduate students their first year in the department.

Teaching Assistant, Caltech Geology, "Earth and the Environment", 2015, 2016, 2017. 12-18 student. 20 hrs/wk

Designed and supervised teaching labs, assisted students with content, oversaw field expeditions.

Research and Work Experience

Postdoctoral Researcher and Lecturer, Dartmouth College EARS, September 2025 – Present

- Researching mineral magnetism of cobalt sulfides with Professor Sarah Slotznick.
- Complementary experimental and modeling investigation of rock and mineral magnetic properties. Mentorship of first year PhD student in lab technique and rock magnetism theory.
- Teaching: EARS 51 *Mineralogy* Winter Term 2026. EARS 2 *Evolution of Earth and Life* Winter Term 2027.
- Undergraduate research mentorship: Early Research Access in Sciences & EARS senior theses.

PhD Candidate in the lab of Prof. Sonia Tikoo, **Stanford University** Geophysics, September 2020 – August 2025

- Thermal modeling of cooling lunar impact ejecta bodies.
- Magnetic field modeling in 3 dimensions in Python.
- Inverse modeling of satellite magnetic field data using MATLAB.
- Paleomagnetism laboratory techniques focused on analysis of extraterrestrial samples.
- Development and implementation of new cutting-edge paleomagnetism laboratory equipment.
- Lab Safety Coordinator responsible for oversight, safety training, and chemical management.

Research Project in the lab of Prof. Laura Schaefer, Stanford University, January 2022-August 2025

- Thermochemical modeling of lunar magma ocean crystallization and trace element partitioning.

GIS Technician, City of Tallahassee, Tallahassee Florida, 2018-2020

- Created and maintained digital databases, maps, and records of city utilities projects.
- Extensive experience with ESRI ArcGIS geospatial platform including software development.

Undergraduate Research Fellow in the lab of Prof. Joe Kirschvink, Caltech, 2015 – 2017

Full-time during summer fellowships and part-time while taking classes.

2017: Extended Late-Cretaceous Magnetostratigraphy of the James Ross Basin Island

- Completed a paleomagnetic study of Antarctic sediment cores for high-precision biostratigraphic dating.

2016: Investigating the Death of the Moyero River Superchron

- Concluded research project on Estonian paleostratigraphy, sample processing and data analysis.
- 2nd author on paper published in the journal Palaeogeography, Palaeoclimatology, Palaeoecology.

2015: Magnetic Dinner Salads: Investigating the Role of Magnetite in Cryopreservation

- Introductory project on paleomagnetic methods with industrial and medical applications.

Publications

Chaffee, Thom, Sonia M. Tikoo, Rachel Maxwell, and Ian Garrick-Bethell (2025). "Testing the Fidelity of Paleopole Determinations From Multidirectionally Magnetized Lunar Crustal Anomaly Source Bodies." *Journal of Geophysical Research: Planets, 130*(9), e2024JE008805

Grappone, J. M., **T. Chaffee**, Y. Isozaki, H. Bauert, and J. L. Kirschvink (2017). "Investigating the duration and termination of the Early Paleozoic Moyero reversed polarity Superchron: Middle Ordovician paleomagnetism from Estonia." *Palaeogeography, Palaeoclimatology, Palaeoecology, 485*: 673-686

In preparation: **Chaffee, Thom**, Zhihua Xiong, Mike J. Krawczynski, and Sarah P. Slotznick. "Mineral magnetism of (cobalt)pentlandite solid solution from synthetic specimens."

In preparation: **Chaffee, Thom**, Hélène Couvy, Mike J. Krawczynski, and Sonia M. Tikoo. "Rock magnetic analysis of schreibersite: temperature and composition dependent parameters."

In preparation: **Chaffee, Thom**, Sonia M. Tikoo, Ji-In Jung, Sam G. Boeschen, Raisha Abubo, and Benjamin P. Weiss. "Null Paleointensities of Young Lunar Impactites."

MAJOR FUNDING

Total: $460,000

Award	Year(s)	PI(s)	My Role	My Approximate Subvention
U. S. Department of Energy Early Career Research Program, Office of Basic Energy Sciences	2025-2027	Sarah P. Slotznick	Postdoctoral lead researcher	$200,000
NASA SSERVI LunaSCOPE	2023-2025	Alex J. Evans	Graduate research fellow	$100,000
NSF Graduate Research Fellowship Program	2020-2025	Thom Chaffee, Sonia M. Tikoo	Graduate research fellow	$160,000

PRESENTATIONS

Chaffee, Thom and Sarah P. Slotznick. “Weak Ferromagnetism of Cobalt Sulfide Minerals.” Presented at 2026 New England Mineral Conference, NEMA, 15-17 May

Chaffee, Thom M., Sonia Tikoo, Hélène Couvy, and Michael J. Krawczynski. "Rock Magnetic Characterization of Natural and Synthetic Schreibersite." GP43B-0363 presented at 2025 Fall Meeting, AGU, 15-19 Dec

Chaffee, Thom. “Impact or Dynamo Fields? Frontiers in Lunar Magnetism.” Invited speaker at Institute of Geophysics and Planetary Physics seminar, University of California Santa Cruz. 2 May 2025

Chaffee, Thom. “No reproducible evidence of impact-generated magnetization in lunar samples.” Invited speaker at Planetary Science and Exploration seminar, Stanford University. 2 April 2025

Chaffee, Thom, S. N. Lamm, S. G. Jarmak, SciX Team. “Planetary Science Data, Object, and Feature Search with the Science Explorer (SciX) Digital Library”. In *56th Lunar and Planetary Science Conference,* no. 3090, p. 2566, March 13 2025

Chaffee, Thom, Sonia M. Tikoo, Sam G. Boeschen, Ji-In Jung, and Benjamin P. Weiss. “Null Paleointensities of 2Ma Lunar Impactite Glass from South Ray Crater”. Presented at LunaSCOPE meeting, August 7 2024

Chaffee, Thom, Sonia M. Tikoo, Rachel Elise Maxwell, and Ian Garrick-Bethell. “Crustal Magnetic Anomalies Preclude a Stable Selenocentric Axial Dipole Structure for the Ancient Lunar Dynamo.” In *55th Lunar and Planetary Science Conference,* no. 3040, p. 2237, March 13 2024

Chaffee, Thom M., Sonia Tikoo, Sam G. Boeschen, Raisha Abubo, Benjamin P. Weiss. “No Evidence of Magnetization from Impact-Generated Fields in 2 Ma Lunar Impact Melt Glasses.” GP24A-08 presented at 2023 Fall Meeting, AGU, 11-15 December

Chaffee, Thom, Sonia Tikoo, Rachel Elise Maxwell, and Ian Garrick-Bethell. “Testing the Robustness of Parker’s Method Against Complexly Magnetized Sources and Implications for Lunar and Planetary Paleopole Determinations”. Bay Area Planetary Science Conference, 19 September 2023

Chaffee, Thom, Sonia M. Tikoo, Raisha Abubo, Sam G. Boeschen, Benjamin P. Weiss. “Testing Whether Lunar Melt Glasses Preserve Records of Impact-Generated Magnetic Fields.” Lunar and Planetary Science Conference, March 2023

Chaffee, Thom M., Sonia Tikoo, Rachel Elise Maxwell, and Ian Garrick-Bethell “Testing the Robustness of Parker’s Method Against Complexly Magnetized Sources and Implications for Lunar and Planetary Paleopole Determinations” GP32B-0350 presented at 2022 Fall Meeting, AGU, 12-16 December

Chaffee, Thom M. and Laura Schaefer. “Oxygen Fugacity in the Lunar Magma Ocean” DI35B-0037 presented at 2022 Fall Meeting, AGU, 12-16 December

Chaffee, Thomas M., and Sonia M. Tikoo. "Size Thresholds for Unidirectional Remanence Within Lunar Magnetic Anomalies." In *52nd Lunar and Planetary Science Conference*, no. 2548, p. 1642. 2021

Chaffee, Thomas M., Ross Mitchell, Sarah P. Slotznick, Jennifer Buz, Joseph Biasi, Joseph O'Rourke, Frank Sousa, David Flannery, Roger R. Fu, and Joseph L. Kirschvink. "Extended Late-Cretaceous Magnetostratigraphy of the James Ross Basin Island, Antarctica." In *AGU Fall Meeting Abstracts*, vol. 2017, pp. GP43A-0972. 2017

Chaffee, Thom M., Joseph L. Kirschvink, and Atsuko K. Kobayashi. "Magnetic Dinner Salads: The Role of Biogenic Magnetite in Cryopreservation for Common Food Plants." In *AGU Fall Meeting Abstracts*, vol. 2015, pp. GP51A-1308. 2015

AWARDS, SERVICE, AND INCLUSIVITY

Awards:

Dartmouth Postdoctoral Association Professional Development Award, July 2026
Funding to attend National Association of Geoscience Teachers Earth Educators’ Rendezvous conference at University of South Carolina.

Dartmouth Earth and Planetary Sciences Elizabeth L. Drake Award for Outstanding Department Service, June 2026
Selected from postdoctoral, research scientist, and lecturer staff by senior faculty in recognition of service

Best paper presentation, New England Mineral Association Conference, May 2026

Institute for Rock Magnetism Visiting Research Fellow
Summer 2026 10-day funded visit to investigate rock magnetism in cobalt sulfide minerals
Winter 2025: 10-day funded visit to investigate rock magnetism in iron phosphide mineral schreibersite

Stanford Centennial Teaching Award, 2024
Nominated by the Geophysics department for commitment to teaching & excellent student feedback.

Lunar and Planetary Science Conference 2023 Lunar and Planetary Institute Career Development Award
For outstanding first-author abstract submission to the 54th Lunar and Planetary Science Conference.

Lunar and Planetary Science Conference 2023 Dwornik Award, Honorable Mention
Award for 2nd-best student presentation at the 53rd Lunar and Planetary Science Conference.

National Science Foundation Graduate Research Fellow (NSF GRFP), 2020-2025
Three years of full PhD funding for a research proposal on lunar magnetic anomalies.

Research Students:

Dankweli Mwaka, Dartmouth senior thesis in Earth and Planetary Sciences, 2026-2027

Keller Bailey, Dartmouth senior thesis in Earth and Planetary Sciences, 2026-2027

Kimia Amiri, Dartmouth undergraduate working in the Slotznick lab under my supervision, 2026-2027
Funded via Dartmouth Early Access in Sciences (ERAS) & Undergraduate Research Assistantships at Dartmouth (URAD)

Mateo Chavez, student at Foothill College, Palo Alto, intern at Stanford Planetary Magnetism Lab under my supervision, Summer 2024

Funded through Foothill Student Learning Institute for first-generation and low income students. Subsequently assisted in successful application for transfer to UCLA & in application for graduate school in Physics.

Sam Boeschen, Stanford undergraduate (coterm) working in the Tikoo lab under my supervision, 2022-2025
Funded through Stanford Introductory Studies and via research credit.
Mentored in developing a novel research project on passive extraction of lunar volatiles for spaceflight, resulting in student-led abstract and presentation at LPSC 2026.

Raisha Abubo, Stanford undergraduate working in the Tikoo lab under my supervision, 2022-2023
Funded through Stanford Introductory Studies.

Ben Otter, Stanford undergraduate working in the Tikoo lab under my supervision for research credit, 2023-2024

Vivian Bahn, high school intern via Stanford Earth Young Investigators program, Summer 2022

Service:

Teacher, Dartmouth Science Day, 2026. 1 day educational event for local K-12 students. Co-ran rock, mineral, and meteorite interactive education station.

Postdoctoral Representative to Faculty Meetings, Dartmouth Earth and Planetary Science, 2025-ongoing

Science Explorer (SciX) Ambassador, 2024-ongoing
Outreach and education role for the digital library and database managed by the Smithsonian Astrophysical Observatory under a NASA cooperative agreement.

Committee on Graduate Studies, Stanford, 2023-2025
Voting member of the faculty senate committee charged with formulating policy on graduate education and reviewing interdisciplinary degree-granting programs.

Stanford Wellness Program Co-Administrator, 2022-2023
Wellness role focused on the larger community for the entire Stanford Doerr School of Sustainability.

Stanford Geophysics Wellness Liaison, 2021-2023
Co-ran department student wellness program, organized student physical and mental health support, workshops for learning wellness skills, and social events.

Inclusion, Diversity, Equity, and Accessibility:

Co-lead, Dartmouth EAPS Inclusive Pedagogy Journal Club, 2026-ongoing
Helped found reading group with Professor Emily Lacroix. Undergraduate, graduate, postdoc, and faculty participants discuss key papers in Earth science education, focus on best practices for inclusivity in the field and classroom.

Participant in Center for Integration of Research, Teaching, and Learning IDEAL Pedagogy course, Stanford, 2024
Ten-week course on using modern, evidence-based pedagogy to meet a diverse range of student needs in the classroom.

Bargaining Representative, Stanford Graduate Workers Union, 2023-2024
Elected labor advocate for LGBTQIA+, disabled, immigrant, and first-generation/low-income students

Participant in student-run “Diversity in Geophysics” course, Stanford, 2020
Project on opening up the hidden curriculum of higher education to first-generation/low-income students.

Accessibility Aide, Caltech, 2017-2018